(7) In a table given below, the values of y are consecutive terms of a series of which 23.6 is the 6th term. Find the first and tenth term of the series.

x	3	4	5	6	7	8	9
y	4.8	8.4	14.5	23.6	36.2	52.8	73.9

Soln:- we need to compute $y(1)$ and $y(10)$.

	x	y	1st Diff	2nd	3rd	4th
x_0	3	y_0 4.8	(3.6) Δy_0	(2.5) $\Delta^2 y_0$	(0.5) $\Delta^3 y_0$	(0) $\Delta^4 y_0$
	4	8.4	6.1	3.0	0.5	0
	5	14.5	9.1	3.5	0.5	0
	6	23.6	12.6	4.0	0.5	0
	7	36.2	16.6	(4.5) $\nabla^2 y_n$	(0.5) $\nabla^3 y_n$	(0) $\nabla^4 y_n$
	8	52.8	(21.1) ∇y_n			
x_n	9	y_n 73.9				

To find $y(1)$, we use Newton-Gregory Forward interpolation formula. Here $x_0 = 3$, $y_0 = 4.8$, $x = 1$, $h = 1$, $P = \frac{x - x_0}{h} = \frac{1-3}{1}$

$$= -2$$

$$y = y_0 + P\Delta y_0 + \frac{P(P-1)}{2!}\Delta^2 y_0 + \frac{P(P-1)(P-2)}{3!}\Delta^3 y_0 + \cdots$$

$$= 4.8 + (-2)(3.6) + \frac{(-2)(-3)}{2}(2.5) + \frac{(-2)(-3)(-4)}{6}(0.5)$$

$$y(1) = 3.1$$

To find $y(10)$, we use Newton-Gregory backward interpolation formula. Here $x_n = 9$, $y_n = 73.9$, $h = 1$, $x = 10$

$$P = \frac{x - x_n}{h} = \frac{10 - 9}{1} = 1$$

$$y = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n + \cdots$$

$$= 73.9 + 1(21.1) + \frac{(1)(2)}{2}(4.5) + \frac{(1)(2)(3)}{6}(0.5) + \cdots$$

$$= 100$$

(8) Find the interpolating polynomial $f(x)$ satisfying $f(0)=0$, $f(2)=4$, $f(4)=56$, $f(6)=204$, $f(8)=496$, $f(10)=980$ and hence find $f(3)$, $f(5)$ and $f(7)$.

Soln: The Difference table is as follows.

x	y	1st Diff	2nd	3rd Diff	4th
x_0 0	y_0 0	(4) Δy_0			
2	4		(48) $\Delta^2 y_0$		
		52		(48) $\Delta^3 y_0$	
4	56		96		0
		148		48	
6	204		144		0
		292		48	
8	496		192		
		484			
10	980				

The interpolating polynomial can be found from either of the two interpolation formula.

We use Newton-Gregory forward interpolation formula.

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \cdots$$

$$p = \frac{x - x_0}{h}, \quad p = \frac{x-0}{2} = \frac{x}{2}$$

$$y = 0 + \frac{x}{2}(4) + \frac{\frac{x}{2}\left(\frac{x}{2}-1\right)}{2}(48) + \frac{\left(\frac{x}{2}\right)\left(\frac{x}{2}-1\right)\left(\frac{x}{2}-2\right)}{6}(48)$$

$$= 2x + \frac{x}{2}\left(\frac{x-2}{2}\right)(24) + \frac{x}{2}\left(\frac{x-2}{2}\right)\left(\frac{x-4}{2}\right)(8)$$

$$= 2x + x(x-2)\,6 + x(x-2)(x-4)$$

$$= x\left[2 + 6(x-2) + (x^2 - 2x - 4x + 8)\right]$$

$$= x\left[2 + 6x - 12 + x^2 - 6x + 8\right]$$

$$= x^3 - 2x$$

Thus the interpolating polynomial is $x^3 - 2x$

Now taking $x = 3, 5, 7$, we obtain

$f(3) = 21$, $f(5) = 115$, $f(7) = 329$.

Interpolation with Unequal intervals

Lagrange's interpolation formula for unequal intervals

$$y = f(x) = \frac{(x-x_1)(x-x_2)\cdots(x-x_n)}{(x_0-x_1)(x_0-x_2)\cdots(x_0-x_n)} y_0$$

$$+ \frac{(x-x_0)(x-x_2)\cdots(x-x_n)}{(x_1-x_0)(x_1-x_2)\cdots(x_1-x_n)} y_1 + \cdots$$

$$+ \frac{(x-x_0)(x-x_1)\cdots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\cdots(x_n-x_{n-1})} y_n$$

(1) Using Lagrange's interpolation find the value of y when $x=2$ from the following data

x	1	3	4	6
y	4	40	85	259

Soln:. Here $x_0=1$ $x_1=3$ $x_2=4$ $x_3=6$

$y_0=4$ $y_1=40$ $y_2=85$ $y_3=259$

Using Lagrange's interpolation formula, we get

$$y = \frac{(x-3)(x-4)(x-6)}{(1-3)(1-4)(1-6)} 4 + \frac{(x-1)(x-4)(x-6)}{(3-1)(3-4)(3-6)} 40$$

$$+ \frac{(x-1)(x-3)(x-6)}{(4-1)(4-3)(4-6)} (85) + \frac{(x-1)(x-3)(x-4)}{(6-1)(6-3)(6-4)} \times 259$$

For $x=2$, we get

$$y = \frac{16}{15} + \frac{160}{3} - \frac{170}{3} + \frac{259}{15}$$

$$y = 15$$

② Using Lagrange's interpolation formula, find f(5) from the following data.

	x_0	x_1	x_2	x_3	x_4
x	1	3	4	6	9
y	−3	9	30	132	156
	y_0	y_1	y_2	y_3	y_4

Soln:- By using Lagrange's interpolation formula

$$y = \frac{(x-3)(x-4)(x-6)(x-9)}{(1-3)(1-4)(1-6)(1-9)}(-3) + \frac{(x-1)(x-4)(x-6)(x-9)}{(3-1)(3-4)(3-6)(3-9)} \times 9$$

$$+ \frac{(x-1)(x-3)(x-6)(x-9)}{(4-1)(4-3)(4-6)(4-9)} \times 30 + \frac{(x-1)(x-3)(x-4)(x-9)}{(6-1)(6-3)(6-4)(6-9)} \times 132$$

$$+ \frac{(x-1)(x-3)(x-4)(x-6)}{(9-1)(9-3)(9-4)(9-6)} \times 156$$

For $x = 5$, we get

$$= -0.1 - 4 + 32 + 46.93 - 1.73$$

$$= 73.1$$

③ The following table gives the normal weights of babies during the first few months of life

	x_0	x_1	x_2	x_3	x_4
Age in Maths	2	5	8	10	12
weight in Kgs	4.4	6.2	6.7	7.5	8.7
	y_0	y_1	y_2	y_3	y_4

Estimate by suitable interpolation method, the normal weight of a baby of 7 months old.

Soln:- Since the data given for x is unequally spaced, we use Lagrange's interpolation formula.

$$y = \frac{(x-5)(x-8)(x-10)(x-12)}{(2-5)(2-8)(2-10)(2-12)} \quad (4.4)$$

$$+ \frac{(x-2)(x-8)(x-10)(x-12)}{(5-2)(5-8)(5-10)(5-12)} \quad (6.2)$$

$$+ \frac{(x-2)(x-5)(x-10)(x-12)}{(8-2)(8-5)(8-10)(8-12)} \quad (6.7)$$

$$+ \frac{(x-2)(x-5)(x-8)(x-12)}{(10-2)(10-5)(10-8)(10-12)} \quad (7.5)$$

$$+ \frac{(x-2)(x-5)(x-8)(x-10)}{(12-2)(12-5)(12-8)(12-10)} \quad (8.7)$$

For $x = 7$, we get

$$y = \frac{(2)(-1)(-3)(-5)}{(-3)(-6)(-8)(-10)} \quad (4.4) + \frac{(5)(-1)(-3)(-5)}{(3)(-3)(-5)(-7)} \quad (6.2)$$

$$+ \frac{(5)(2)(-3)(-5)}{(6)(3)(-2)(-4)} \quad (6.7) + \frac{(5)(2)(-1)(-5)}{(8)(5)(2)(-2)} \quad (7.5)$$

$$+ \frac{(5)(2)(-1)(-3)}{(10)(7)(4)(2)} \quad (8.7)$$

$$= -0.09 + 1.48 + 6.98 - 2.34 + 0.47$$

$$= 6.5 \text{ kg}$$

Inverse interpolation

The process of estimating the value of x for a given value of y is called the inverse interpolation.

Lagrange's inverse interpolation formula

$$x = \frac{(y-y_1)(y-y_2)\cdots(y-y_n)}{(y_0-y_1)(y_0-y_2)\cdots(y_0-y_n)} x_0$$

$$+ \frac{(y-y_0)(y-y_2)\cdots(y-y_n)}{(y_1-y_0)(y_1-y_2)\cdots(y_1-y_n)} x_1$$

$$+ \cdots + \frac{(y-y_0)(y-y_1)\cdots(y-y_{n-1})}{(y_n-y_0)(y_n-y_1)\cdots(y_n-y_{n-1})} x_n$$

(1) Given

x	1	3	4
y	4	12	19

find x corresponding to $y=7$.

OR

If $y_1=4$, $y_3=12$, $y_4=19$ and $y_x=7$ find x?

Soln. Here $x_0=1$ $x_1=3$ $x_2=4$

$y_0=4$ $y_1=12$ $y_2=19$

By Lagrange's inverse interpolation formula, we have

$$x = \frac{(y-12)(y-19)}{(4-12)(4-19)}(1) + \frac{(y-4)(y-19)}{(12-4)(12-19)}(3) + \frac{(y-4)(y-12)}{(19-4)(19-12)}\times 4$$

For $y=7$, $x = \frac{60}{120} + \frac{108}{56} - \frac{60}{105}$

$= 0.5 + 1.9285 - 0.5714 = 1.8571$

(2) Apply Lagrange's method to find the value of x corresponding to $f(x)=15$ from the following data.

	x_0	x_1	x_2	x_3
x	5	6	9	11
y	12	13	14	16
	y_0	y_1	y_2	y_3

Soln: By Lagrange's inverse interpolation formula with $y=f(x)=15$.

$$x=\frac{(15-13)(15-14)(15-16)}{(12-13)(12-14)(12-16)}\times 5+\frac{(15-12)(15-14)(15-16)}{(13-12)(13-14)(13-16)}\times 16$$

$$+\frac{(15-12)(15-13)(15-16)}{(14-12)(14-13)(14-16)}\times 9+\frac{(15-12)(15-13)(15-14)}{(16-12)(16-13)(16-14)}\times 11$$

$$=\frac{10}{8}-\frac{18}{3}+\frac{54}{4}+\frac{66}{24}$$

$$=11.5$$

Numerical Differentiation

The process of evaluating the derivative of a function at some particular value of the independent variable, when a set of given values of the function is known is called Numerical Differentiation.

Formula:- (using forward Difference)

$$\frac{dy}{dx} = \frac{1}{h}\left[\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 - \frac{1}{4}\Delta^4 y_0 + \cdots\right]$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12}\Delta^4 y_0 - \frac{5}{6}\Delta^5 y_0 + \cdots\right]$$

Formula: (using backward Difference)

$$\frac{dy}{dx} = \frac{1}{h}\left[\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n + \frac{1}{4}\nabla^4 y_n + \cdots\right]$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \cdots\right]$$

(1) Given

x	1.0	1.2	1.4	1.6	1.8	2.0
y	2.72	3.32	4.06	4.96	6.05	7.39

find y' and y'' at $x = 1.2$.

Soln:- To compute y' and y'' we take $x_0 = 1.2$

Since the value is near the beginning of the table we use forward difference formula.

x	y	1st Diff	2nd Diff	3rd Diff	4th Diff
1.0	2.72				
		0.60			
x_0 1.2	y_0 3.32		0.14		
		(0.74) Δy_0		0.02	
1.4	4.06		(0.16) $\Delta^2 y_0$		0.01
		0.90		(0.03) $\Delta^3 y_0$	
1.6	4.96		0.19		(0.03) $\Delta^4 y_0$
		1.09		0.06	
1.8	6.05		0.25		
		1.34			
2.0	7.39				

Here $h=0.2$, $\Delta y_0 = 0.74$, $\Delta^2 y_0 = 0.16$, $\Delta^3 y_0 = 0.03$, $\Delta^4 y_0 = 0.03$

$$\frac{dy}{dx} = \frac{1}{h}\left[\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 - \frac{1}{4}\Delta^4 y_0 + \cdots\right]$$

$$= \frac{1}{0.2}\left[0.74 - \frac{0.16}{2} + \frac{0.03}{3} - \frac{0.03}{4}\right]$$

$$= 3.3125.$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12}\Delta^4 y_0\right]$$

$$= \frac{1}{(0.2)^2}\left[0.16 - 0.03 + \frac{11}{12}(0.03)\right]$$

$$= 3.9375.$$

(2) The following table gives the values of $\sin\theta$ for different values of θ:

θ	0°	10°	20°	30°	40°
$\sin\theta$	0.0000	0.1736	0.3420	0.5000	0.6428

Find the value of $\cos 10^\circ$.

Soln: Here $h=10$. To compute $\cos 10^\circ$ we have to find $\frac{d}{d\theta}(\sin\theta)$. we take $x_0 = 10^\circ$

Soln:

	$\theta =$	$\sin\theta$	1st Diff	2nd Diff	3rd Diff	4th Diff
	0	0.0000				
			0.1736			
x_0	10	$y_0 = 0.1736$	Δy_0	−0.0052		
			0.1684	$\Delta^2 y_0$	−0.0052	
	20	0.3420		−0.0104		0.0004
			0.1580		−0.0048	
	30	0.5000		−0.0152	$\Delta^3 y_0$	
			0.1428			
	40	0.6428				

Note $10^\circ = \frac{\pi}{18}$ radians

$$= \frac{1}{h}\left[\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 - \cdots\right]$$

$$\left.\frac{d}{d\theta}(\sin\theta)\right|_{\theta=\frac{18}{\pi}} = \frac{18}{\pi}\left[0.1684 - \frac{1}{2}(-0.0104) + \frac{1}{3}(-0.0048)\right]$$

$$= \frac{18}{\pi}[0.1684 + 0.0052 - 0.0016]$$

$$= \frac{18}{\pi}[0.172]$$

$$= 0.9854$$

(3) Find $f'(2.2)$ and $f''(2.2)$ given the following table

x	1.4	1.6	1.8	2.0	2.2
$f(x)$	4.0552	4.9530	6.0496	7.3891	9.0250

Soln: Here $h = 0.2$, $x_n = 2.2$, The Difference table is given by

x	y	1st Diff	2nd Diff	3rd Diff	4th Diff
1.4	4.0552				
		0.8978			
1.6	4.9530		0.1988		
		1.0966		0.0441	
1.8	6.0496		0.2429		0.0094
		1.3395		0.0535	
2.0	7.3891		0.2964		
		1.6359			
x_n 2.2	y_n 9.0250				

∇y_n $\nabla^2 y_n$ $\nabla^3 y_n$ $\nabla^4 y_n$

$$\frac{dy}{dx} = \frac{1}{h}\left[\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n + \frac{1}{4}\nabla^4 y_n + \cdots\right]$$

$$= \frac{1}{0.2}\left[1.6359 + \frac{1}{2}(0.2964) + \frac{1}{3}(0.0535) + \frac{1}{4}(0.0094)\right]$$

$$= 5\left[1.6359 + 0.1482 + 0.0178 + 0.0023\right]$$

$$= 9.0212$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \cdots\right]$$

$$= \frac{1}{(0.2)^2}\left[0.2964 + 0.0535 + \frac{11}{12}(0.0094)\right]$$

$$= 25\left[0.2964 + 0.0535 + 0.0086\right]$$

$$= 8.9625$$

(4) In a machine a slider moves along a fixed straight rod. Its distance x (cm) along the rod for various values of time t (sec) are given below. Find the velocity and acceleration of the slider at $t = 0.3$ sec.

t	0	0.1	0.2	0.3	0.4	0.5
x	29.91	31.62	32.87	33.64	33.95	33.81

Soln: The Difference table is

t	x	1st Diff	2nd Diff	3rd Diff	4th Diff	5th Diff
0	29.91					
		1.71				
0.1	31.62		−0.46			
		1.25		$\nabla^3 y_n$		
0.2	32.87		−0.48 $\nabla^2 y_n$	−0.02		
		0.77 ∇y_n			0.04	
0.3	y_n 33.64		−0.46	0.02		−0.05
		0.31			−0.01	
0.4	33.95		−0.45	0.01		
		−0.14				
0.5	33.81					

Here $h=0.1$, $t_n=0.3$

→ Since the dependent variable in this problem is "x"

$$\frac{dx}{dt} = \frac{1}{h}\left[\nabla x_n + \frac{1}{2}\nabla^2 x_n + \frac{1}{3}\nabla^3 x_n + \frac{1}{4}\nabla^4 x_n + \cdots\right]$$

$$= \frac{1}{0.1}\left[0.77 + \frac{1}{2}(-0.48) + \frac{1}{3}(-0.02)\right]$$

$$= 10\,[0.77 - 0.24 - 0.0066]$$

Velocity $= 5.233$ cm/sec

$$\frac{d^2x}{dt^2} = \frac{1}{h^2}\left[\nabla^2 x_n + \nabla^3 x_n + \frac{11}{12}\nabla^4 x_n + \cdots\right]$$

$$= \frac{1}{(0.1)^2}[-0.48 - 0.02]$$

acceleration $= -50$ cm/sec^2

(5) Find $\frac{dy}{dx}$ at $x=1$ from the following table of values

x	1	2	3	4
y	1	8	27	64

Soln: The Difference table is given by

x	y	1st Diff	2nd Diff	3rd Diff
x_0 1	y_0 1	(7) Δy_0	(12) $\Delta^2 y_0$	(6) $\Delta^3 y_0$
2	8	19	18	
3	27	37		
4	64			

we have $h=1$, $x_0=1$,

$$\frac{dy}{dx} = \frac{1}{h}\left[\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 - \cdots\right]$$

$$= \frac{1}{1}\left[7 - \frac{1}{2}(12) + \frac{1}{3}(6) - \cdots\right]$$

$$= 3$$